

watering.ino

```

1 #include<NTPClient.h>
2 #include<ESP8266WiFi.h>
3 #include<WiFiUdp.h>
4 #include<OneWire.h>
5 #include<DallasTemperature.h>
6 #include<DFRobot_DHT11.h>
7 #include<DHT.h>
8 const char* ssid = "*****";
9 const char* pass = "*****";
10 const long utcOffsetInSeconds = 19800; // IST = UTC +5:30 that is [(5*60)+30]*60 sec
11 WiFiUDP ntpUDP;
12 NTPClient timeClient(ntpUDP, "pool.ntp.org", utcOffsetInSeconds);
13 DFRobot_DHT11 DHT;
14 #define DHT11_PIN D2 // pin D2 for DHT11
15 const int oneWireBus = D3; // pin D3 for ds18b20
16 int temp;
17 OneWire oneWire(oneWireBus);
18 DallasTemperature sensors(&oneWire);
19 void setup()
20 {
21   Serial.begin(115200);
22   pinMode(D2, INPUT); // DHT11

```

Fig. 5: Code snippet Wi-Fi configuration

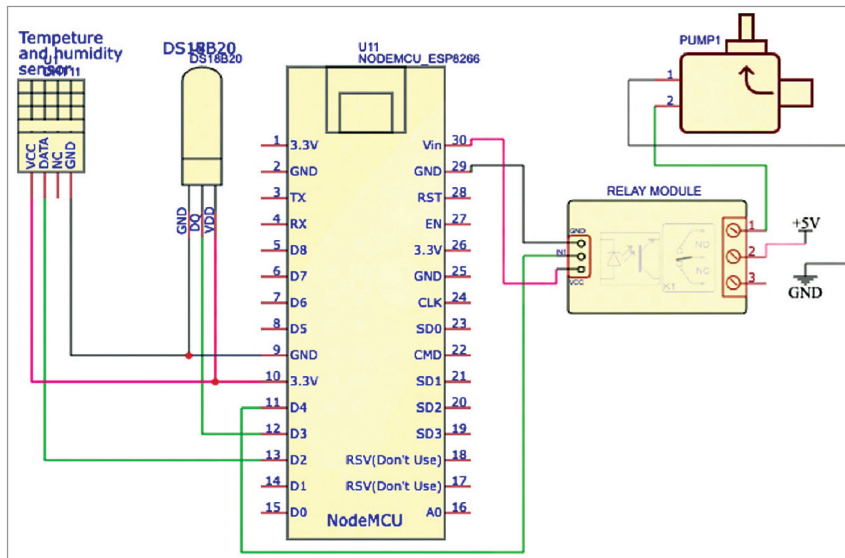


Fig. 6: Circuit diagram of the project

without damage even when running dry. These pumps can achieve outlet pressures of up to 15 Bar and are suitable for handling aggressive and viscous fluids. Maintenance is easy, as only the hose, which comes into contact with the fluid, needs to be replaced. Peristaltic pumps also feature reversible flow and heavy-duty bearings greased for life.

Every setup has unique requirements, and in the present setup, a significant flow of water through the misting nozzle and sufficient throw

to reach the plants housed in the upper and lower bars of the hydroponics system was desired. To measure the atmospheric temperature and humidity, we used the DHT11 sensor. The DS18B20 sensor was employed to measure the temperature of the water used in the hydroponics setup. The 4-way fogger operates based on the time of the day, with 4 pm being

the ideal time when the temperature starts to subside as the sun moves westward. Providing an extra shower at this time would provide relief to the indoor plants. Hence, certain modifications were introduced to the existing commercially available set, as illustrated in Fig. 4. The figure also depicts the setup arrangements made for the operation.

The components required to build this setup are listed in Table 1.

Software

The entire system is programmed using Arduino IDE. However, please note that the Arduino IDE does not officially support ESP boards. You can add the ESP board by following the instructions provided in the link: [\[https://arduino-esp8266.readthedocs.io/en/latest/installing.html\]](https://arduino-esp8266.readthedocs.io/en/latest/installing.html) (<https://arduino-esp8266.readthedocs.io/en/latest/installing.html>)

After adding the ESP board, install the following libraries using the library manager in Arduino IDE: DallasTemperature, DFRobot_DHT11, and NTPClient.

In the code, you need to include the libraries, define the pins for sensors, and configure the Wi-Fi according to your network's SSID and password. Fig. 5 showcases a code snippet for Wi-Fi configuration.

Next, you need to create the setup function where you initialise the serial port for debugging, set up the Wi-Fi network, check the connection status, and establish communication with the sensors. Following that, you need to create the loop function. Here, you check the sensor data and time using NTP, and based on the time, you turn the water valve pin on and off. Make sure to select the NODE MCU ESP8266 as the board and then upload the code.

Circuit diagram

Fig. 6 displays the circuit diagram of the proposed project setup. It is built

EFY Note

The source code of this project is available for free download at www.electronicsforu.com